

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders With Flow Designer

Team ID	SWTID-2026-1668
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Team Members:

Name	Role
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2. Project Overview

Purpose: Standard laptop requests in many organisations are handled through emails, calls and spreadsheets, which makes them slow and hard to track. This project automates the whole process on the ServiceNow platform using Flow Designer, so that a request is validated, approved, fulfilled and communicated without manual chasing.

Goals:

- Reduce the time taken to process standard laptop requests.
- Give employees one simple form to request a laptop and follow its status.
- Enforce a consistent approval and procurement process with a clear audit trail.

Features:

- Service Catalog item for standard laptop requests with required variables.
- Automatic manager approval routed by Flow Designer.
- Automatic creation of a fulfilment task for the procurement team after approval.
- Automatic request status updates and email notifications to the requester.
- Handling of rejected requests, with the outcome communicated to the requester.
- Flow execution history for monitoring and troubleshooting.

Note on the template: this template is written for MERN projects. This project is built on ServiceNow, so the sections below keep the same headings and describe the equivalent ServiceNow components (catalog and portal in place of React, Flow Designer in place of Node.js/Express, and ServiceNow tables in place of MongoDB).

3. Architecture

Frontend: Instead of a React application, users interact through the ServiceNow Service Catalog / Service Portal. A catalog item named for the standard laptop request presents a form with variables (for

example, requester details, laptop model and justification). Submitting the form creates a request and requested item (RITM) record.

Backend: Instead of Node.js and Express.js, the server-side logic runs in ServiceNow Flow Designer. A flow is triggered by the catalog request and performs these steps in order: read request data, ask the manager for approval, create a catalog task for the procurement group, update the request state, and send notifications.

Step	Flow component	Purpose
1	Trigger: Service Catalog	Starts the flow when a laptop request is submitted
2	Action: Ask for Approval	Sends the request to the requester's manager
3	Flow logic: If / Else	Branches on approved or rejected
4	Action: Create Catalog Task	Assigns fulfilment to the procurement group
5	Action: Update Record	Sets request/RITM state as the process progresses
6	Action: Send Email	Notifies the requester of each outcome

Database: Instead of MongoDB, data is stored in ServiceNow's relational platform tables. No custom schema is required for the core process.

Table	Name	Used for
sc_request	Request	Parent request raised by the employee
sc_req_item	Requested Item (RITM)	The laptop item and its variable values
sc_task	Catalog Task	Fulfilment work for the procurement team
sysapproval_approver	Approval	Manager approval decision
sys_user / sys_user_group	User / Group	Requesters, managers and procurement group

4. Setup Instructions

Prerequisites:

- A ServiceNow instance (a free Personal Developer Instance from developer.servicenow.com is sufficient).
- An administrator account on the instance.
- A web browser (Chrome, Edge or Firefox).
- Sample users: a requester, a manager and a procurement group member, with the manager set on the requester's user record.
- Outbound email configured on the instance if email notifications are to be delivered.
- Git, only if you wish to clone the project repository.

Installation:

1. Clone the repository: `git clone https://github.com/karthikar-12/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-With-Flow-Designer`
2. Log in to your ServiceNow instance as an administrator.
3. Create the required users and groups (requester, manager, procurement group) and assign the needed roles.

4. Import the project configuration (for example, the update set provided in the repository) through System Update Sets > Retrieved Update Sets, then preview and commit it. If no update set is used, create the catalog item and flow manually as described in section 3.
5. Open Flow Designer, find the laptop request flow, and confirm it is published (Active).
6. Confirm the catalog item is published in the Service Catalog category available to users.

Environment variables: none are required. Instance URL and credentials are handled by ServiceNow; if the REST API is used from an external client, keep the credentials outside the code.

5. Folder Structure

ServiceNow stores configuration in the instance rather than in client/server folders. The project is organised logically as follows.

Client (user-facing):

Component	Description
Catalog item	Standard laptop request form and its variables
Catalog category	Places the item where employees can find it
Portal page / widgets	Service Portal pages used to request and track items

Server (logic):

Component	Description
Flow (Flow Designer)	Main flow with trigger, approval, task creation, update and email actions
Notifications	Email templates sent on submission, approval/rejection and completion
Users, groups and roles	Access control for requesters, approvers and fulfillers
Update set / repository files	Exported configuration and documentation stored in GitHub

6. Running the Application

There are no local frontend and backend servers to start; the application runs on the ServiceNow instance.

1. Frontend: open the Service Portal / Service Catalog of the instance and log in as the requester.
2. Submit the standard laptop request form.
3. Backend: open Flow Designer > Executions to watch the flow run, log in as the manager to approve, then as a procurement user to complete the catalog task.

7. API Documentation

The solution is driven by catalog forms and Flow Designer and does not expose custom endpoints. The standard ServiceNow REST APIs can be used to interact with the same records. Replace {instance} with your instance name.

Method	Endpoint	Description
GET	/api/now/table/sc_req_item	List requested items. Params: sysparm_query, sysparm_limit
GET	/api/now/table/sc_req_item/{sys_id}	Get one requested item by sys_id
GET	/api/now/table/sc_task	List catalog tasks. Params: sysparm_query

Method	Endpoint	Description
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Order a catalog item with variable values in the body
PATCH	/api/now/table/sc_task/{sys_id}	Update a catalog task (for example, its state)

Example request (order_now body):

```
{ "sysparm_quantity": "1", "variables": { "laptop_model": "Standard", "justification": "New joiner" } }
```

Example response (shape, abbreviated):

```
{ "result": { "sys_id": "...", "number": "REQ0010001", "request_number": "REQ0010001", "request_id": "..." } }
```

Variable names in the body must match the variable names defined on the catalog item in your instance.

8. Authentication

- Users log in to ServiceNow with their own accounts (local credentials, or SSO if configured by the organisation).
- Sessions are managed by the platform after login; no custom tokens are implemented in this project.
- REST API calls use HTTP Basic authentication or OAuth 2.0 bearer tokens, as configured on the instance.
- Authorization is handled by roles, groups and Access Control Lists (ACLs): requesters can create and view their own requests, managers can approve, and only the procurement group can work on catalog tasks.

9. User Interface

The user interface is the ServiceNow Service Catalog / Service Portal. Insert screenshots of the following screens.

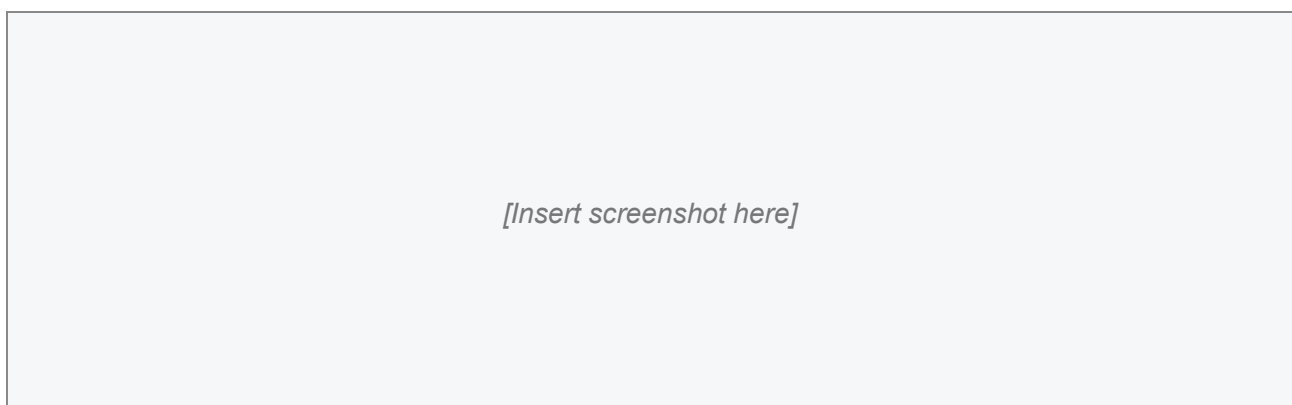


Figure 1: Standard laptop request catalog item form

[Insert screenshot here]

Figure 2: Request and requested item status view

10. Testing

Testing follows a manual, scenario-based strategy and user acceptance testing (UAT), using the flow execution history to verify each step. The results are summarised in the separate UAT report.

Scenario	Expected result
Submit a request with all required fields	Request and RITM are created; requester is notified
Submit with a mandatory field missing	Form blocks submission and highlights the field
Manager approves	Flow continues and a catalog task is created for procurement
Manager rejects	Request is closed as rejected; requester is notified
Procurement completes the task	Request is marked complete; requester is notified
Review Flow Designer execution details	Every step is logged with inputs and outputs

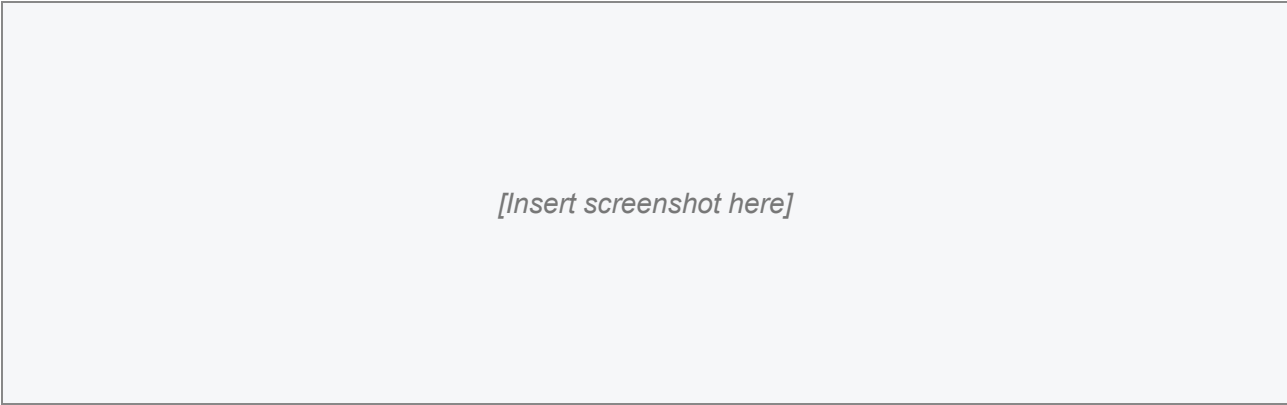
Tools: ServiceNow Flow Designer test/execution tools, catalog item preview, and impersonation of the requester, manager and fulfiller roles.

11. Screenshots or Demo

Demo Link	https://drive.google.com/file/d/1JE4BSQaxmTknBbU6lXir0fwZvhYc4nJJ/view?usp=drivesdk
Document Link	https://drive.google.com/file/d/1fd2vbx21ivYQ8ffxW2LJMr7I-ucZmzf5/view?usp=drivesdk
GitHub	https://github.com/karthikar-12/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-With-Flow-Designer

[Insert screenshot here]

Figure 3: Flow in Flow Designer (trigger, approval, task and notification actions)



[Insert screenshot here]

Figure 4: Catalog task created for the procurement team

12. Known Issues

- Only standard laptop orders are covered; non-standard or custom hardware needs a separate process.
- Approvals depend on a correct manager being set on each user record; otherwise the approval cannot be routed.
- Email notifications are delivered only when outbound email is configured on the instance.
- A free Personal Developer Instance may hibernate after inactivity and needs to be woken before use.
- There is no integration yet with external vendor or inventory systems.

13. Future Enhancements

- Extend the flow to other standard assets such as monitors, phones and accessories.
- Integrate with vendor systems for automatic purchase orders and delivery tracking.
- Add inventory checks so available stock is allocated before a new purchase is raised.
- Add multi-level approvals by cost or department, and SLA tracking with escalations.
- Build dashboards and reports for request volume, approval time and fulfilment time.